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WAGHOLI, PUNE

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Minor Degree in Cyber Security
TAE-2
Subject: Fundamental of Cyber Security & Cryptography

Mini Project Report On
"Sentinel: Next-Generation Firewall & Intrusion Detection System"

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Academic Year 2026-27
Semester-V

CERTIFICATE

This is to certify that the mini project entitled **“Sentinel: Next-Generation Firewall & Intrusion Detection System”** embodies the original work done by **AMMAR RAISAHMAD MULLA** as a partial fulfilment of the requirement of TAE-2 for the subject **Fundamental of Cyber Security & Cryptography** under the Minor Degree in Cyber Security, Third Year B. Tech., Semester V, of Pune University during the academic year 2026-2027.

Date: 01/10/2026
Place: Pune

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Prof. Birudev Yele**, my project guide, for his valuable guidance, encouragement, and constant technical support throughout the development of this mini project on the **Sentinel: Next-Generation Firewall & Intrusion Detection System**. His insights into network defense paradigms, packet filtering rulesets, and heuristic intrusion detection helped shape both the implementation and this report.

I am deeply thankful to **Mrs. Sarita Patil**, Head of the Department of Computer Engineering, for providing the necessary computing laboratories and an encouraging academic environment. I also thank all the staff members of the Department of Computer Engineering for their assistance whenever required.

Finally, I extend my heartfelt appreciation to my family and friends, and to everyone who supported me directly or indirectly in the completion of this project.

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1. Problem Definition & Objectives

Problem Definition

Modern enterprise computing infrastructure and cloud networks are subjected to persistent, automated cyber-attacks. These range from volumetric Distributed Denial of Service (DDoS) vectors—such as TCP SYN floods designed to saturate server transmission control block (TCB) memory queues—to automated horizontal and vertical port scanning recon probes seeking vulnerabilities across critical administrative daemons (SSH port 22, RDP port 3389, and SMB port 445). Traditional enterprise firewall utilities, including Linux iptables, operate as opaque black boxes providing static text logs that lack real-time interactive telemetry and visual threat trajectory mapping.

Objectives

- **Layer 4 Stateful Packet Inspection:** To develop a firewall engine evaluating incoming packets against configured rules (IP, port, protocol) with real-time ALLOW/DROP verdicts.
- **Heuristic Intrusion Detection (IDS):** To implement an anomaly engine that flags malicious vectors like TCP SYN Floods (MITRE ATT&CK; T1498.001) and port scans.
- **Interactive Draggable Topology Mesh:** To engineer a dynamic topology graph using SVG math and cubic Bézier curves, providing visual representation of threats and internal LAN assets.
- **Adversary Attack Simulation Console:** To create a simulation console enabling security teams to trigger live attack scenarios on demand to validate packet dropping.
- **Dual Operational Modes:** To support both 'Live Network Mode' (sniffing host TCP sockets via Server-Sent Events) and 'Simulation Mode'.
- **SecOps AI Forensic Analyst:** To integrate an intelligent assistant capable of root-cause analysis on flagged attacker IPs with 1-click drop rules.
- **Relational Database Persistence:** To store firewall rules and logs in MySQL ('firewall_db') with persistent hit and byte counters.

2. Introduction & Theoretical Background

The Sentinel Firewall & Intrusion Detection System is an advanced SecOps platform designed to combine stateful packet filtering, heuristic threat signature matching, live socket streaming, and interactive topology modeling. This chapter covers the theoretical principles underpinning next-generation firewall architectures.

2.1 Next-Generation Firewalls (NGFW) vs Traditional Packet Filters

Traditional packet filters operate statelessly at Layers 3 and 4, inspecting IP headers in isolation. Next-Generation Firewalls (NGFWs) integrate stateful inspection, connection state tracking (SYN, SYN-ACK, ESTABLISHED), and application awareness. Sentinel combines stateful policy evaluation with heuristic anomaly detection.

2.2 Common Network Attack Vectors

- **TCP SYN Flood DDoS (MITRE T1498.001):** Exploits the three-way handshake by transmitting rapid SYN frames without completing handshakes, overflowing kernel connection tables.
- **Port Reconnaissance (MITRE T1046):** Scanning common ports (22, 80, 443, 445, 3389) to enumerate vulnerable services.
- **Credential Brute-Force:** Repeated connection bursts against remote access daemons (SSH/RDP).

Table 1 : Traditional Firewalls vs Sentinel Next-Gen Firewall

Parameter	Traditional Packet Filter	Sentinel Next-Gen Firewall
Inspection Layer	Layer 3 & 4 only	Layer 3, 4 & Heuristic Signatures
Connection State	Stateless	Stateful + Velocity Tracking
Intrusion Detection	None (separate daemon)	Built-in Heuristic IDS Engine
Attack Simulation	Requires external tools	Integrated Live Simulation Console

Topology Modeling	Textual CLI iptables	Interactive Draggable Mesh (Bézier curves)
Database Sync	Flat text files	Relational MySQL database
AI Forensics	None	Sentinel AI Analyst (Gemini / DeepSeek)

3. System Architecture & Data Pipelines

Sentinel follows a decoupled three-tier architecture ensuring sub-millisecond evaluation latency, browser rendering stability, and persistent database state across sessions.

3.1 Architectural Tiers

• **Tier 1: Presentation Layer (React 19 + Tailwind CSS):** Renders the Live Terminal, Draggable Topology Mesh, IP Geo Radar, and AI Copilot.
• **Tier 2: Core Engine Layer (App Runtime):** Executes the FIFO Queue Dispatcher, Netfilter Policy Matcher, Heuristic IDS, and Telemetry Calculators.
• **Tier 3: Backend Bridge & Persistence (Node.js + MySQL):** Sniffs host sockets, broadcasts SSE streams, and updates 'firewall_db'.

Table 3 : Core System Modules

Module	Input	Processing Logic	Output
Live Terminal	Packet stream	Rule matching & formatting	Monospace log stream
Netfilter Engine	IP, Port, Protocol	Hierarchical rule evaluation	ALLOW/DROP verdict
Heuristic IDS	Live packets	Velocity & signature inspection	Threat feed & alerts
Topology Mesh	Origin & LAN nodes	Pointer dragging & Bézier curves	Interactive network mesh
Attack Console	Attack vector config	High-rate SYN packet flood	DDoS load & spike
AI Analyst	Forensic queries	Live telemetry + Gemini API	Root-cause analysis

4. Implementation: Design Web Pages & UI Modules

The user interface is engineered as a high-contrast, tactical Security Operations Center (SOC) dashboard. The figures below document the implemented views alongside screenshot reference placeholders.

4.1 Command Center Dashboard (Home View)

■ **[SCREENSHOT 2: Sentinel Command Center Dashboard]**
Shows top telemetry cards (Throughput Mbps, PPS, Allow/Drop counters), dynamic sparkline chart, and traffic mode toggle.

4.2 Live Traffic Terminal with Continuous Sequential Stream

■ **[SCREENSHOT 3: Live Traffic Terminal Feed]**
Shows packets streaming one-by-one with timestamps, source/destination IPs, protocols, and green [ALLOW] / red [DROP] tags.

4.3 Network Topology Mesh Graph (Interactive Draggable Nodes)

■ **[SCREENSHOT 4: Interactive Topology Mesh Graph]**
Shows the central Sentinel Firewall Core, connected attacker nodes (red dashed curves), allowed transit nodes (green curves), and internal LAN servers.

4.4 Live Attack Simulation Console & Critical Banner

■ **[SCREENSHOT 5: Attack Simulation Console & Threat Banner]**
Shows the live attack console executing a TCP SYN Flood DDoS from 192.168.1.105 with top red Critical Alert Banner.

4.5 Intrusion Detection System (IDS) Threat Feed

■ **[SCREENSHOT 6: IDS Threat Intelligence Feed]**
Displays the live-updating detected security events table with attack vectors, confidence ratings, and 1-click Auto-Block buttons.

4.6 IP Geolocation Threat Radar Map

■ **[SCREENSHOT 7: IP Geolocation Threat Radar Map]**
Shows the world map with trajectory arcs and origin nodes table featuring locations such as Mumbai, India and Frankfurt, Germany.

4.7 Sentinel AI Analyst & Forensic Chatbot

■ **[SCREENSHOT 8: Sentinel AI Analyst Copilot]**
Shows the interactive forensic chat drawer with root-cause explanations and 1-click 'Commit DROP Rule' buttons.

5. Algorithms & Working Mechanism

5.1 Netfilter Rule Matching Algorithm

Sentinel sorts all active policies by specificity: explicit IP matches precede wildcard '0.0.0.0/0', and specific ports precede 'ALL'. The first rule satisfying IP, port, and transport protocol constraints determines the packet verdict (ALLOW or DROP).

5.2 Dynamic Cubic Bézier Curves for Topology Edges

For smooth, responsive edge rendering as nodes are dragged, cubic Bézier curves are computed between node coordinates P0(x0,y0) and P3(x3,y3) using intermediate control points P1(x0 + (x3-x0)/2, y0) and P2(x0 + (x3-x0)/2, y3). The equation guarantees horizontal tangents:

$B(t) = (1-t)^3 P0 + 3(1-t)^2 t P1 + 3(1-t) t^2 P2 + t^3 P3$

5.3 Sequential FIFO Queue Dispatcher

To eliminate packet batch bursts when sniffing active OS sockets, Sentinel queues incoming packets in an in-memory buffer (packetQueueRef) and dispatches exactly one packet every 500ms–600ms, maintaining a continuous, natural stream.

```
// Pseudocode: Netfilter Packet Evaluation & Heuristic IDS Inspection
function evaluatePacketAgainstRules(packet, rules):
sortedRules = sortRulesBySpecificity(rules)
for r in sortedRules:
if match(packet.srcIp, r.ip) and match(packet.dstPort, r.port) and match(packet.proto, r.proto):
return { action: r.action, matchedRule: r.id }
return { action: 'ALLOW', matchedRule: 'DEFAULT' }

function inspectThreatHeuristics(packet):
if packet.proto == 'TCP [SYN]' and packet.dstPort == 443:
return triggerAlert('TCP SYN Flood (DDoS)', 'ET-DOS-SYNFLOOD', 'CRITICAL')
if packet.dstPort in [22, 3389] and packet.rate > THRESHOLD:
return triggerAlert('Brute-Force Probe', 'ET-AUTH-PROBE', 'HIGH')
```

6. Technology Stack & Deployment Architecture

Table 5 : Technology Stack

Component	Technology	Role & Purpose
Frontend UI	React 19, Tailwind CSS 4	Reactive SOC dashboard and SVG topology
Bundler	Vite 8.3	Fast dev server and module bundling
Backend Server	Node.js 22, Express 5	Native socket sniffer and SSE stream
Database	MySQL 8.0 (mysql2)	Rule persistence and audit logging
AI Copilot	Google Gemini / DeepSeek	Forensic reasoning and root cause analysis

7. Results, Verification & Testing

Sentinel was tested across multiple network conditions, adversary attack simulations, and database persistence operations. All test cases passed successfully as documented in Table 7.

Table 7 : Verification Test Cases & Empirical Results

TC#	Scenario	Input	Expected Output	Status
TC-01	Admin Auth	Ammar / hackAMR	Access granted	PASS
TC-02	Invalid Login	Wrong password	Access denied	PASS
TC-03	Terminal Stream	Normal baseline	Continuous 1-by-1 flow	PASS
TC-04	SYN Flood Simulation	Trigger attack	Speedup + Alert Banner	PASS
TC-05	DROP Rule Enforcement	Block 192.168.1.105	Packets marked [DROP]	PASS
TC-06	Topology Node Drag	Pointer drag node	Smooth Bézier curves follow	PASS
TC-07	Topology Layout Reset	Click Reset Layout	Default positions restored	PASS
TC-08	MySQL Persistence	Reload page	Rules reload with hits	PASS
TC-09	IP Geo Radar	Track Mumbai IP	Country badge [IN] visible	PASS
TC-10	AI Copilot Query	Forensic prompt	Analysis returned	PASS

8. Conclusion & Future Scope

The Sentinel: Next-Generation Firewall & Intrusion Detection System project successfully demonstrates how stateful packet filtering, heuristic signature anomaly detection, and interactive topology modeling can be synthesized into an educational, production-ready SecOps platform. By providing real-time visibility into packet trajectories, automated attack mitigation, and persistent database accounting, the project strengthened our understanding of confidentiality, integrity, and availability within the Minor Degree in Cyber Security curriculum.

Future Scope

- **Kernel-Level eBPF Probing:** Transition packet interception to Linux eBPF for sub-microsecond in-kernel dropping.
- **Deep Learning Anomaly Detection:** Integrate neural Autoencoder models to identify zero-day polymorphic traffic.
- **Multi-Node Cluster Mesh:** Synchronize distributed firewall policies across multiple cloud VPCs via Raft consensus.

9. References & Project Repositories

- NIST Special Publication 800-41 Rev. 1, 'Guidelines on Firewalls and Firewall Policy,' 2020.
- J. Postel, 'Transmission Control Protocol (TCP),' RFC 793, IETF, 1981.
- W. Eddy, 'TCP SYN Flooding Attacks and Common Mitigations,' RFC 4987, IETF, 2007.
- MITRE Corporation, 'Direct Network Flood (T1498.001),' MITRE ATT&CK; Framework, 2024.
- William Stallings, 'Cryptography and Network Security: Principles and Practice,' 8th Edition, Pearson.

GitHub Repository: https://github.com/ammarcraft/sentinel-firewall
Local Runtime: http://localhost:5174/ (Frontend) | http://localhost:5000/ (Backend API)